

V3 empirical study · Instruction

Two models · four underlyings · four volatility regimes · 1.5-year monthly calibration · option-implied Heston fami

What this report is

- Companion to the seven-model 10,000-path V4 study. This report keeps only Heston and Heston–Merton. In V4 their P dynamics still come from returns (Method A), and listed calls identify the Pan (2002) volatility / jump-size premia used for LSM. Rankings are within this pair; they are not compared here with return-based models.
- Companies are SPY, AAPL, MSFT, and AMZN. No new calibration or LSM.
- Return-based companion: V4_1p5y_monthly_empirical_study_return_based.pdf.
- Computational source of truth: the companion Jupyter notebook. This PDF is written from the same payload; every table number is identical.
- Question: which of 2 American-call pricing models tracks listed market prices most closely, and does the ranking change across companies and volatility regimes?
- In every table the BEST row is the lowest percentage RMSE. That row is bold and shaded green, and the Ranking column ranks 1 = best.

Experimental design (held fixed for every model)

- Models: Heston (no jumps), Heston–Merton (Bates).
- Companies: SPY (primary), AAPL, MSFT, AMZN. Each name is calibrated and priced separately.
- Regimes (V3 one-year evaluation windows): Crisis 2008-08-01→2009-07-31; Normal 2014-01-01→2014-12-31; Late-cycle 2018-10-01→2019-09-30; COVID 2019-09-01→2020-08-31.
- The late-cycle and COVID files share September 2019 (V3 notebook dates). All other regime days are disjoint. Contracts are built independently in each window.
- Calibration window: 1.5-year ending at each monthly update. Rolling: monthly (period start plus month-ends). Parameters estimated at t are used only after t (no look-ahead).
- If history is shorter than the requested lookback, the estimator uses all available observations up to the update date (equity starts 2003-12-01; listed-option panels start 2008-01).

V3 empirical study · Instruction (continued)

Two models · four underlyings · four volatility regimes · 1.5-year monthly calibration · option-implied Heston fami

Shared market sample and filters (not model-specific)

- For each company $\times$ regime, one listed-call sample is drawn once and reused by all two models in this report: nearest-ATM call on each Monday (next session if Monday is closed), DTE 7–60, listed expiry, as-of that date.
- Estimation filters, applied in order to every quote used for Heston/Heston–Merton calibration and to the LSM comparison set: (1) calls only, finite S , K , C ; (2) no-arbitrage $C \geq \max(0, S-K)$; (3) $7 \leq \text{DTE} \leq 60$; (4) $|S/K - 1| \leq 10\%$; (5) premium ≥ 0.05 , valid bid-ask with relative spread $\leq 50\%$ when those columns exist, volume ≥ 1 when present.
- Return-based models (GBM, Merton, GARCH, GARCH–Merton) never see option quotes in §4; they still price the same filtered LSM contracts in evaluation.

Model-correctness adjustments (V3)

- Heston / Heston–Merton: $(\kappa, \theta, \xi, \rho, v_0)$ are option-implied Fourier NLS (Albrecher little-trap), not Method A realized-variance moments. μ is the P-measure lookback mean of stock returns; LSM replaces $\mu \rightarrow r$.
- Euler ordering: the stock increment over $[t, t+\Delta t]$ uses the current variance v_t ; $v_{t+\Delta t}$ is updated afterwards. No look-ahead in the variance.
- LSM: Longstaff–Schwartz American call on the full path cloud ($n_{\text{paths}} = 10000$, seed 42, $n_{\text{steps}} = \text{DTE}$, $\Delta t = 1/252$). Paths are not averaged before stopping.

Metrics

- Primary: $\text{RMSE\%} = 100 \times \sqrt{\text{mean}(((C_{\text{model}} - C_{\text{mkt}}) / C_{\text{mkt}})^2)}$. Ranking uses this number only.
- Secondary: $\text{MAE\%} = 100 \times \text{mean}(|C_{\text{model}} - C_{\text{mkt}}|/C_{\text{mkt}})$. $\text{Bias\%} = 100 \times \text{mean}((C_{\text{model}} - C_{\text{mkt}})/C_{\text{mkt}})$. Positive bias = model expensive vs market.

Page 3 · Filters, sample, and no-look-ahead

Shared evaluation sample · n listed calls (identical for all two models in this report)

Regime	Window	SPY n	AAPL n	MSFT n	AMZN n
2008-2009 Crisis	2008-08-01 → 2009-07-31	50	50	50	50
2013-2014 Normal	2014-01-01 → 2014-12-31	50	50	50	50
2018-2019 Late-cycle	2018-10-01 → 2019-09-30	50	51	50	51
2019-2020 COVID	2019-09-01 → 2020-08-31	50	50	50	50

Filter funnel · last-step n in the 1.5-year lookback ending at each regime (processed call panel)

Ticker	Regime	Input*	No-arb	DTE 7-60	S/K-1 ≤10%	Liquidity
SPY	2008-2009	3359	3311	3311	3203	3203
SPY	2013-2014	12815	12548	12548	12405	12405
SPY	2018-2019	35222	35104	35104	34809	34809
SPY	2019-2020	42100	41961	41961	41379	41379
AAPL	2008-2009	685	685	685	652	652
AAPL	2013-2014	8925	8875	8875	8633	8633
AAPL	2018-2019	6309	6297	6297	6084	6084
AAPL	2019-2020	8010	7992	7992	7713	7713
MSFT	2008-2009	566	566	566	538	538
MSFT	2013-2014	3299	3237	3237	3175	3175
MSFT	2018-2019	7814	7752	7752	7583	7583
MSFT	2019-2020	6892	6852	6852	6655	6655
AMZN	2008-2009	465	465	465	442	442
AMZN	2013-2014	5715	5692	5692	5598	5598
AMZN	2018-2019	28809	28808	28808	28346	28346
AMZN	2019-2020	26766	26760	26760	26323	26323

*Input is the processed call panel restricted to (regime end – 1.5-year, regime end]. Processed files already drop many raw quotes; the funnel above is the additional V3 research filter.

No look-ahead. Monthly parameters at update date t use only equity returns and listed quotes with trading_date $\leq t$. LSM prices the Monday contract using the latest calibration row with date $\leq$ that Monday. Quotes are as-of the grid timestamp (same session, else the prior session).

SPY 2008–2009 uses bid-ask mid when the source mark is \$0.01 (a GitHub field error, not missing trades). After that repair the processed panel has weekly coverage, so n matches AAPL/MSFT (~50). All two models in this report still share that exact sample.

Heston / Heston-Merton calibration further subsamples the filtered lookback surface to ≤ 24 contracts (maturity $\times$ moneyness strata). GBM / Merton / GARCH / GARCH-Merton estimate from lookback log returns only (jumps: 3σ residual threshold where applicable).

Page 4 · Table block · SPY · two models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table SPY-1 SPY · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
Heston	25.90	22.28	+13.29	1
Heston-Merton	38.05	33.61	+25.36	2

Table SPY-2 SPY · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
Heston	59.26	47.06	+44.22	1
Heston-Merton	74.75	66.23	+65.36	2

Table SPY-3 SPY · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
Heston	49.97	41.27	+36.18	1
Heston-Merton	70.07	59.97	+58.12	2

Table SPY-4 SPY · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
Heston	47.54	40.64	+29.57	1
Heston-Merton	72.56	65.60	+56.44	2

Page 5 · Table block · Company summary · two models × four regimes

Each number is the arithmetic mean of AAPL, MSFT, AMZN. Same columns as the SPY page. Bold green row = lowest mean RMSE%. 1.5-year lookahead, rolling=monthly, n_paths=

Table Co-1 Company summary · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · mean of AAPL, MSFT, AMZN · $\bar{n} = 50$

Model	RMSE%	MAE%	Bias%	Ranking
Heston	27.88	23.64	+15.57	1
Heston-Merton	31.27	26.93	+20.08	2

Table Co-2 Company summary · 2013-2014 Normal
2014-01-01 → 2014-12-31 · mean of AAPL, MSFT, AMZN · $\bar{n} = 50$

Model	RMSE%	MAE%	Bias%	Ranking
Heston	35.79	28.32	+26.44	1
Heston-Merton	42.25	34.77	+33.43	2

Table Co-3 Company summary · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · mean of AAPL, MSFT, AMZN · $\bar{n} = 51$

Model	RMSE%	MAE%	Bias%	Ranking
Heston	30.54	24.32	+19.61	1
Heston-Merton	39.87	33.71	+32.00	2

Table Co-4 Company summary · 2019-2020 COVID
2019-09-01 → 2020-08-31 · mean of AAPL, MSFT, AMZN · $\bar{n} = 50$

Model	RMSE%	MAE%	Bias%	Ranking
Heston	34.55	28.93	+20.82	1
Heston-Merton	43.42	37.74	+31.56	2

RMSE%, MAE%, and Bias% are each averaged across AAPL, MSFT, AMZN. This is not a pooled re-pricing of a combined contract sample. Ranking uses the mean RMSE% only.

Page 6 · Table block · AAPL · two models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table AAPL-1 AAPL · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
Heston	31.07	25.81	+19.95	1
Heston-Merton	33.89	28.49	+23.08	2

Table AAPL-2 AAPL · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
Heston	34.83	27.66	+27.17	1
Heston-Merton	40.21	33.41	+33.36	2

Table AAPL-3 AAPL · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 51 contracts

Model	RMSE%	MAE%	Bias%	Ranking
Heston	34.04	28.08	+25.17	1
Heston-Merton	42.20	35.75	+34.85	2

Table AAPL-4 AAPL · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
Heston	34.22	28.97	+23.49	1
Heston-Merton	41.76	36.21	+31.48	2

Page 7 · Table block · MSFT · two models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table MSFT-1 MSFT · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
Heston	25.98	23.13	+9.45	1
Heston-Merton	29.00	26.08	+15.28	2

Table MSFT-2 MSFT · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
Heston	40.25	33.10	+32.34	1
Heston-Merton	50.19	42.58	+42.11	2

Table MSFT-3 MSFT · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
Heston	27.61	21.05	+15.81	1
Heston-Merton	40.75	34.60	+32.94	2

Table MSFT-4 MSFT · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
Heston	33.07	26.92	+19.27	1
Heston-Merton	43.27	37.93	+32.57	2

Page 8 · Table block · AMZN · two models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=monthly, n_paths=10000, seed=42.

Table AMZN-1 AMZN · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
Heston	26.60	21.97	+17.33	1
Heston-Merton	30.92	26.21	+21.89	2

Table AMZN-2 AMZN · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
Heston	32.30	24.22	+19.81	1
Heston-Merton	36.36	28.31	+24.81	2

Table AMZN-3 AMZN · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 51 contracts

Model	RMSE%	MAE%	Bias%	Ranking
Heston	29.98	23.82	+17.85	1
Heston-Merton	36.66	30.79	+28.23	2

Table AMZN-4 AMZN · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE%	Bias%	Ranking
Heston	36.37	30.91	+19.68	1
Heston-Merton	45.25	39.09	+30.63	2

Page 7 · Summary tables · RMSE% across companies and regimes

Table S1 · Overall ranking (mean RMSE% over 16 cells)

Rank	Model	Mean RMSE%	Median RMSE%	# of 16 cells best
1	Heston	35.56	33.55	16
2	Heston-Merton	45.37	41.25	0

Table S2 · Wins by company (lowest RMSE% in that company×regime)

Model	SPY wins	AAPL wins	MSFT wins	AMZN wins	Total
Heston	4	4	4	4	16
Heston-Merton	0	0	0	0	0

Table S3 · Mean RMSE% by regime (equal-weight mean of SPY / AAPL / MSFT / AMZN)

Model	2008-2009 Crisis	2013-2014 Normal	2018-2019 Late-cycle	2019-2020 COVID	Mean of 4
Heston	27.39	41.66	35.40	37.80	35.56
Heston-Merton	32.96	50.38	47.42	50.71	45.37

Green row in S1 / S3 = lowest mean RMSE%. Green row in S2 = most company×regime wins. A model can win the most cells without having the lowest average error (and vice versa).

Table S4.1 · SPY

Model	Mean RMSE%	Median	# regimes best
Heston	45.67	48.75	4
Heston-Merton	63.86	71.32	0

Table S4.2 · AAPL

Model	Mean RMSE%	Median	# regimes best
Heston	33.54	34.13	4
Heston-Merton	39.51	40.98	0

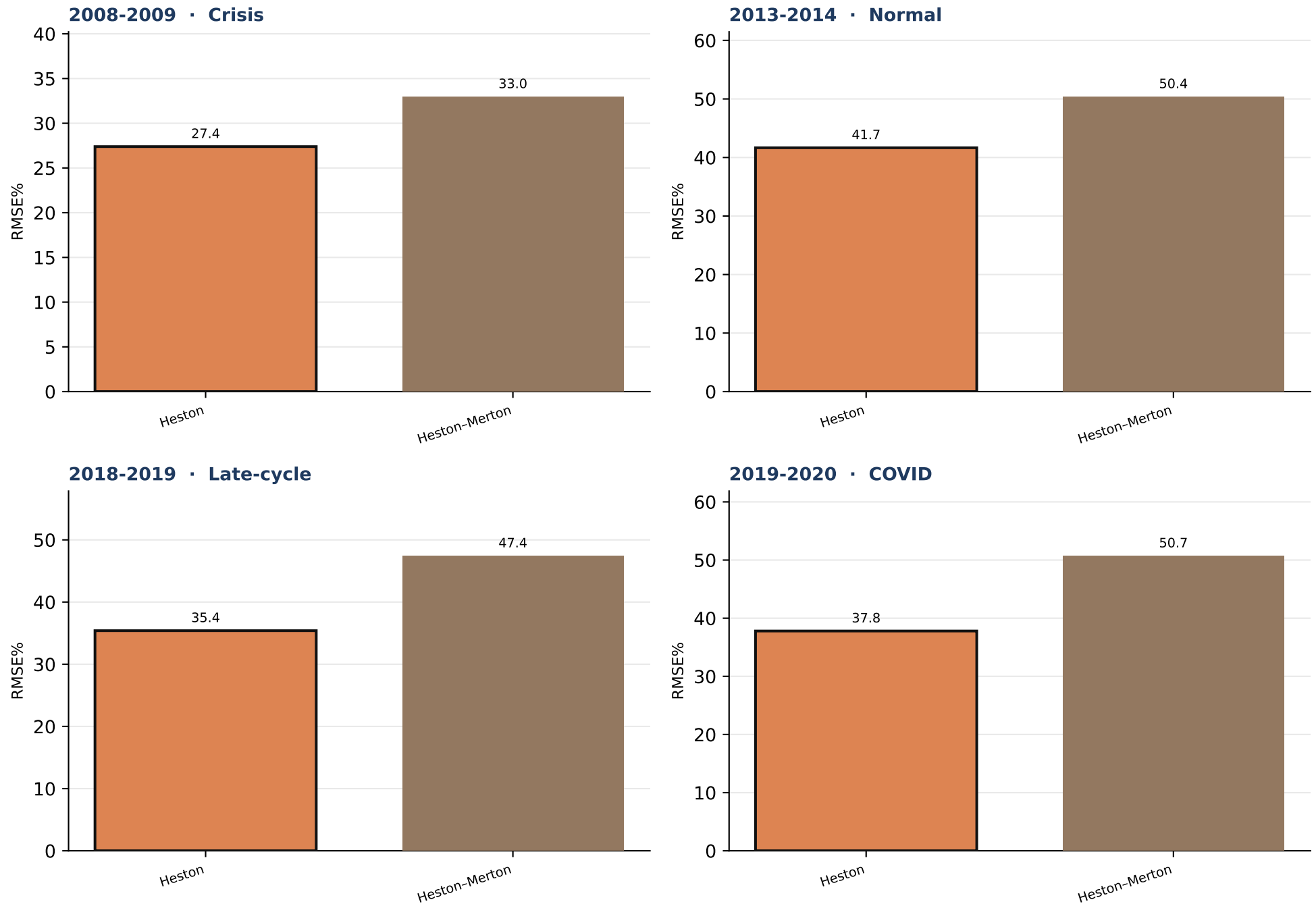
Table S4.3 · MSFT

Model	Mean RMSE%	Median	# regimes best
Heston	31.73	30.34	4
Heston-Merton	40.80	42.01	0

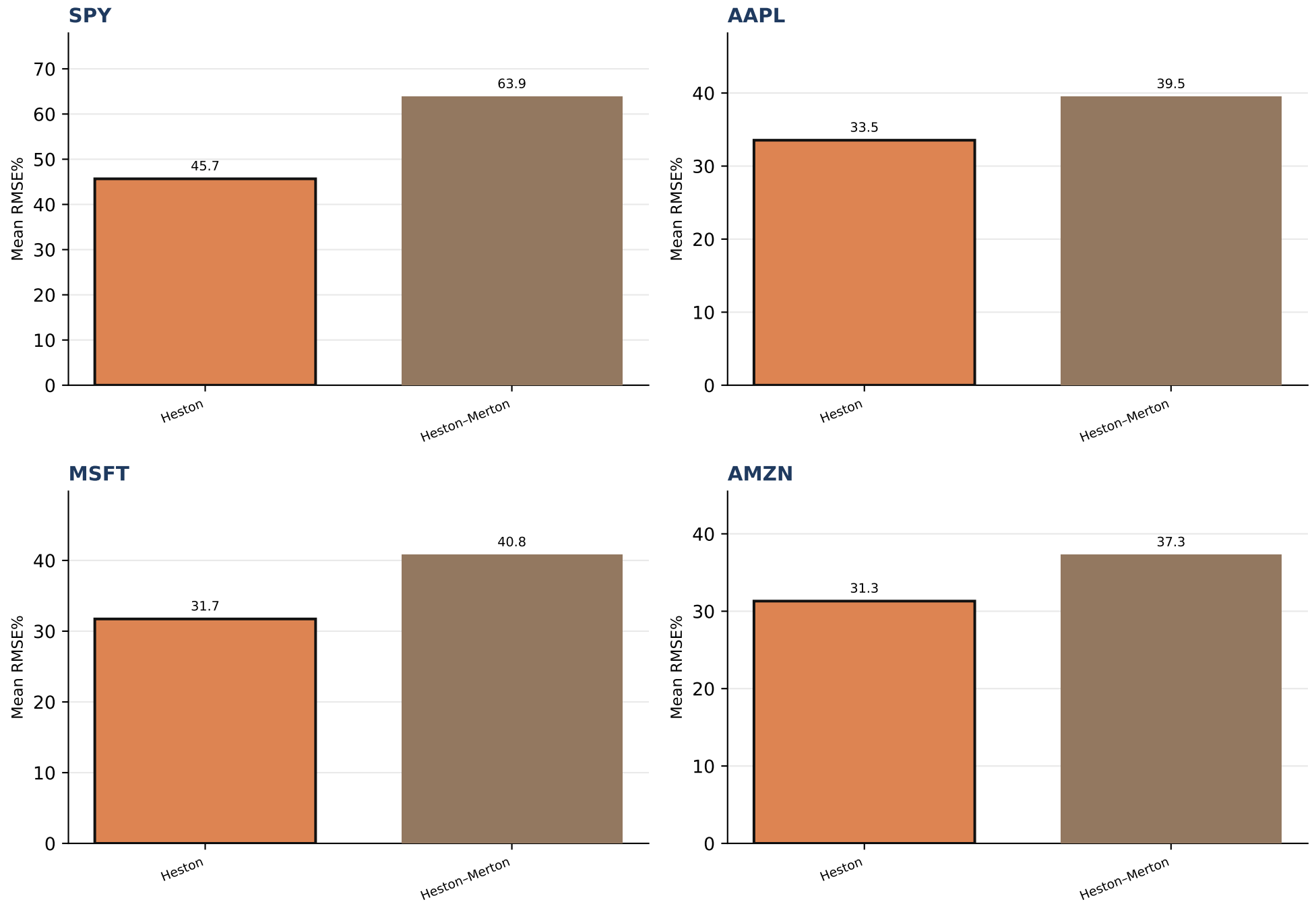
Table S4.4 · AMZN

Model	Mean RMSE%	Median	# regimes best
Heston	31.31	31.14	4
Heston-Merton	37.30	36.51	0

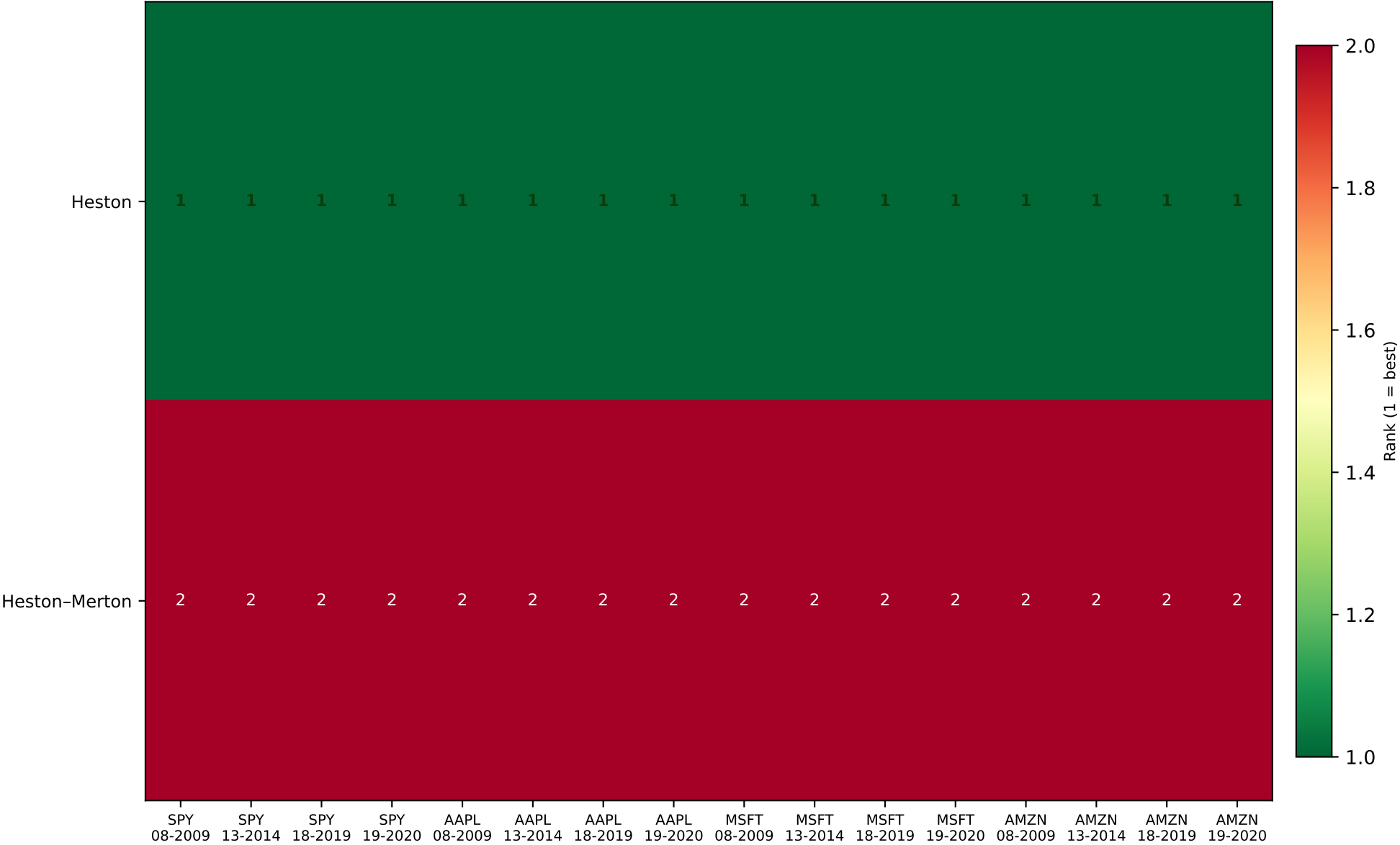
Each panel is one company. Green row = lowest mean RMSE% across the four V3 regimes for that name.



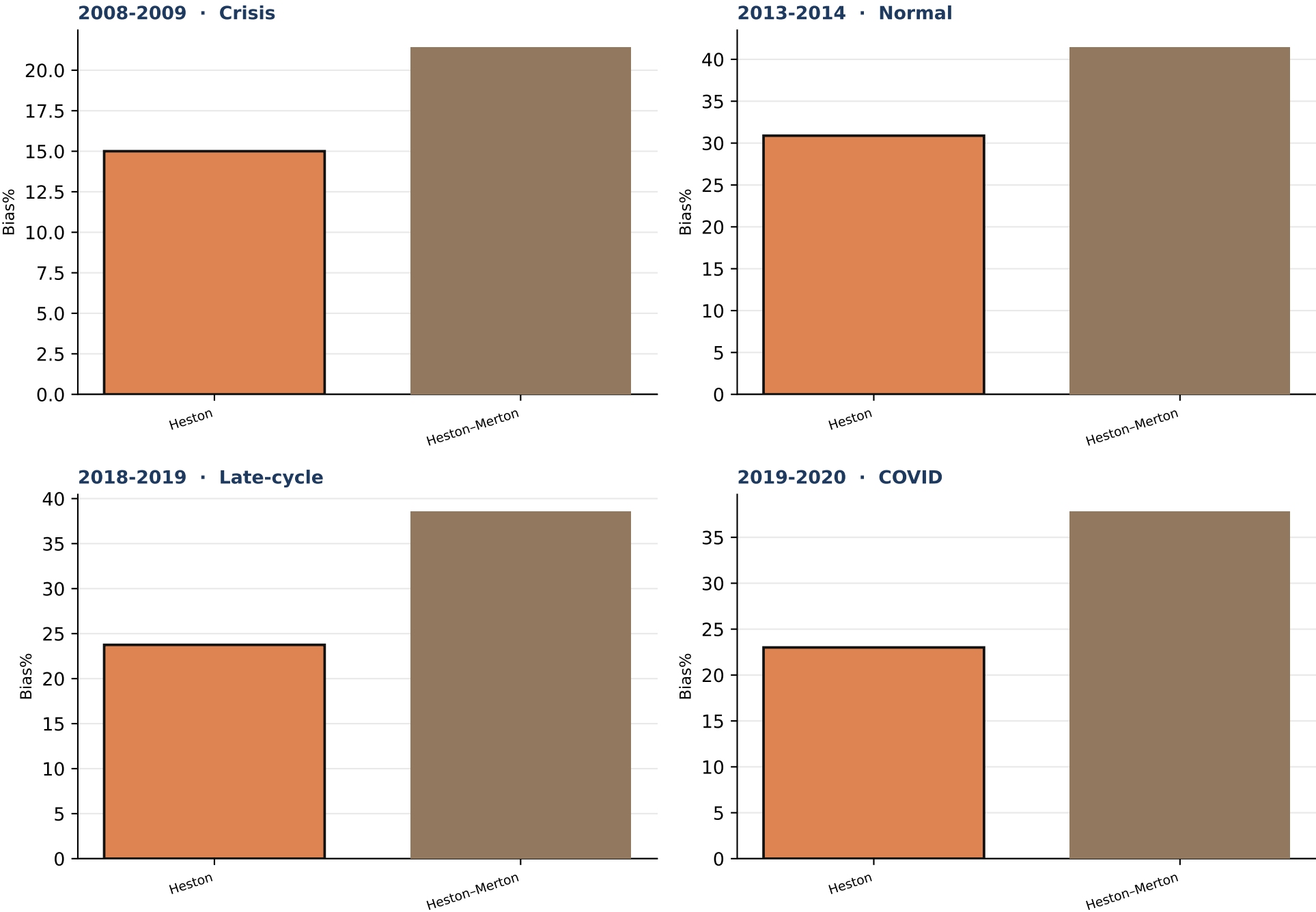
Outlined bar = lowest mean RMSE% in that regime. Equal-weight mean of the four companies.



Outlined bar = lowest mean RMSE% for that company.



Green / 1 = lowest RMSE% in that company×regime cell. Rankings can and do change across names and volatility states.



Outlined bar = closest to zero Bias% (not the RMSE ranking). Positive = model expensive vs listed quote.

Which models perform best, and does the ranking change?

- On mean percentage RMSE across the 16 company×regime cells, Heston is the best overall model (mean RMSE% = 35.56; 16 of 16 cells).
- Second overall is Heston-Merton (mean RMSE% = 45.37; 0 cells).
- The ranking is stable: Heston is best in every company and every regime.
- SPY: lowest mean RMSE% is Heston (45.67); wins 4 of 4 regimes.
- AAPL: lowest mean RMSE% is Heston (33.54); wins 4 of 4 regimes.
- MSFT: lowest mean RMSE% is Heston (31.73); wins 4 of 4 regimes.
- AMZN: lowest mean RMSE% is Heston (31.31); wins 4 of 4 regimes.
- 2008-2009 (Crisis): best mean RMSE% is Heston (27.39).
- 2013-2014 (Normal): best mean RMSE% is Heston (41.66).
- 2018-2019 (Late-cycle): best mean RMSE% is Heston (35.40).
- 2019-2020 (COVID): best mean RMSE% is Heston (37.80).
- No-jump models as a group have lower mean RMSE% (35.56) than the jump models in this report (45.37).
- For the overall winner (Heston), crisis RMSE% is 27.39 vs 41.66 in the 2014 normal year — absolute pricing error is regime-dependent even when the ranking is not.
- These conclusions are conditional on the V3 filters, 1.5-year monthly calibration, Monday ATM sample, LSM with 10,000 paths, and American-call quotes only. They are not a statement about European options or other tenors.

How to read the 12 detailed tables

- Each of Tables SPY-1...MSFT-4 uses exactly the same option contracts for all two models in this report. Differences are the dynamics, not the sample.
- BEST is always lowest RMSE%. MAE% and Bias% are reported, not used for the ranking.
- A model that is BEST on RMSE% can still be biased (systematically expensive or cheap).